

Data Management and File Structures-I

- In this presentation, two exercises and a homework with four questions are given for you to work on their solutions to prepare for the exam(s).
- The homework has to be handed in the beginning of the week.

EXERCISE 1

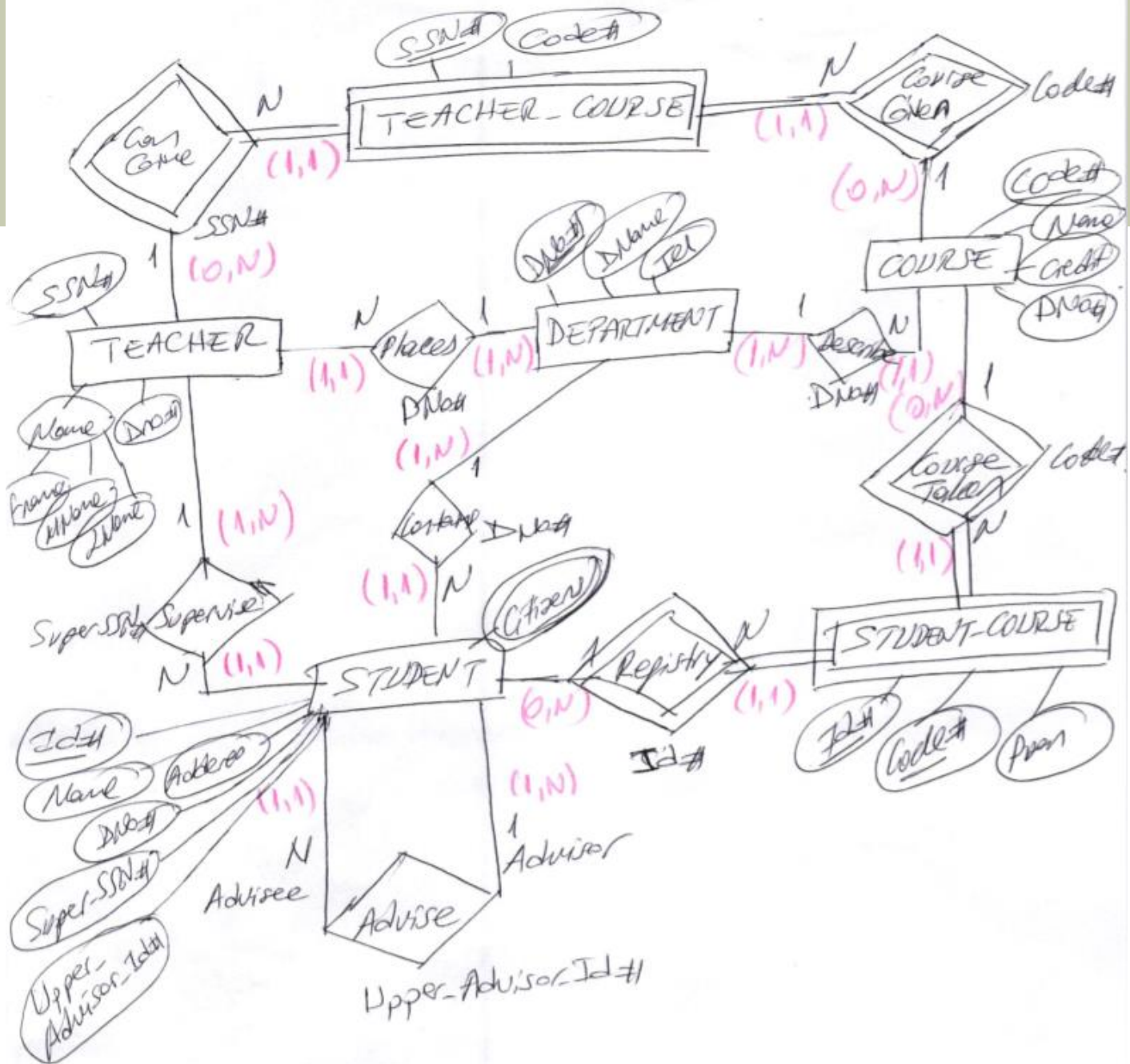
Relational database schema (based on Relational Model) is given below:

DEPARTMENT (DNo#, DName, Tel, ---)
TEACHER (SSN#, Name (FName, MName, LName), DNo#)
COURSE (Code#, Name, Credit, DNo#)
TEACHER_COURSE (SSN#, Code#)
STUDENT (Id#, Name, Address, Tel, DNo#, Super-SSN#)
 {Citizens}, Upper-Advisor-Id#)
STUDENT_COURSE (Id#, Code#, Puan)

- Student and Teacher can only be the members of a department.
- Supervisor and advisor can have one or more student(s).

Draw the diagram of the model showing relations, relationship types, weak relations and relationtypes.

SOLUTION 1

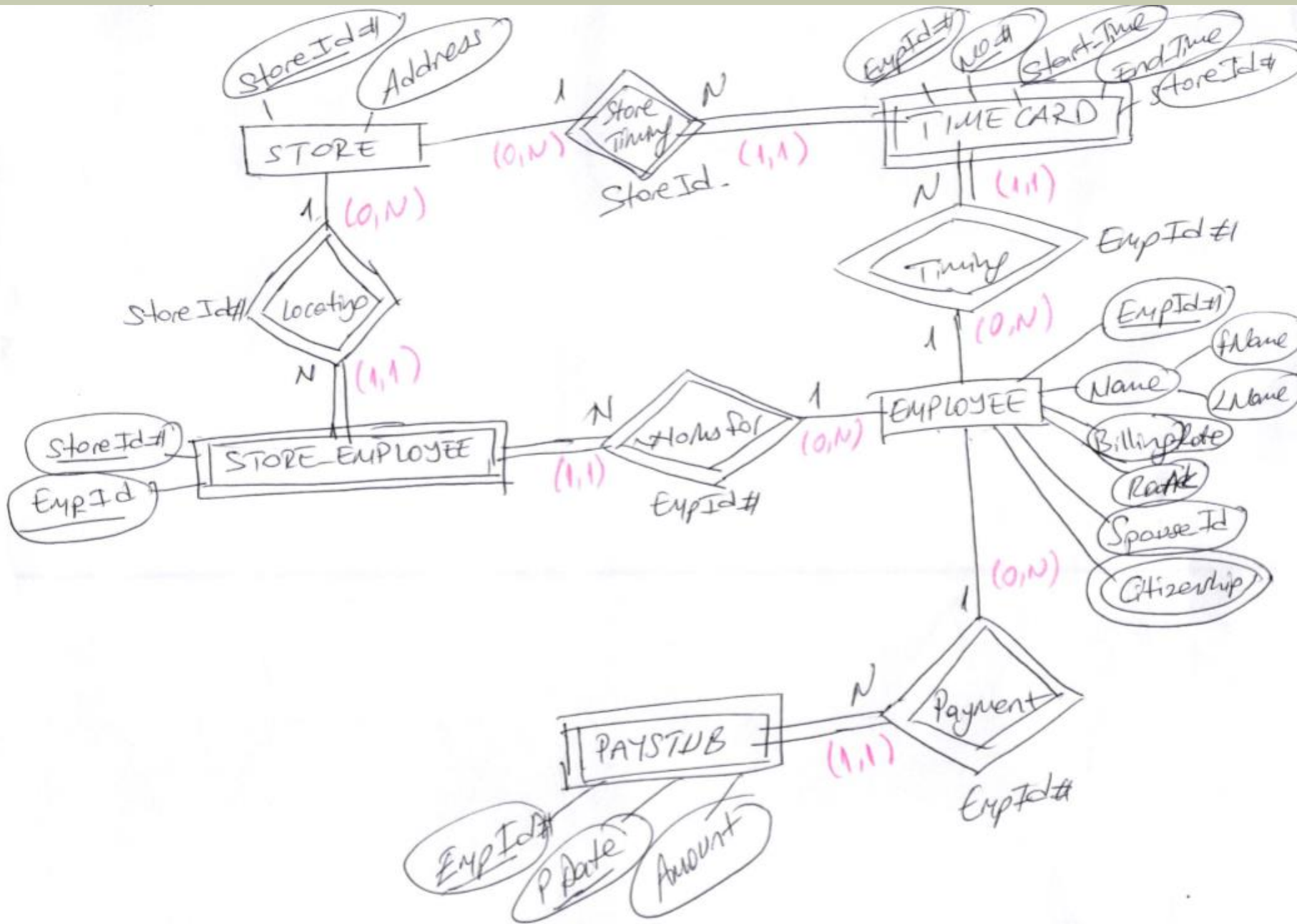


EXERCISE 2

Relational database schema (based on Relational Model) is given below:

STORE (StoreId#, Address)
STORE-EMPLOYEE (StoreId#, EmpId#)
EMPLOYEE (EmpId#, Name (fName, lName), Billing Rate, Rank,
Spouse Id, {Citizenship})
TIMECARD (EmpId#, No#, Start Time, End Time, Store Id)
PAYSTUB (EmpId#, Date, Amount)

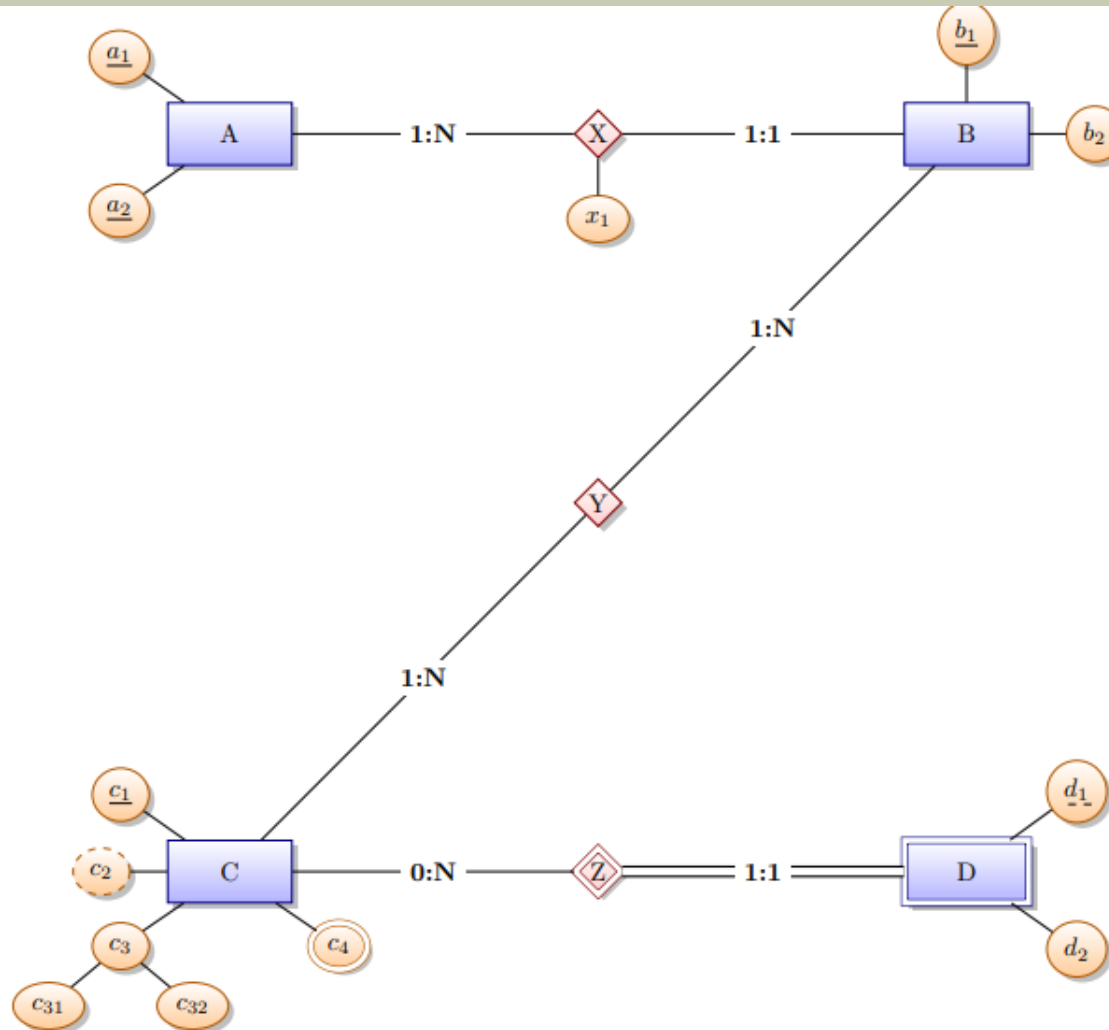
Draw the diagram of the model showing relations, relationship types, weak relations and relationtypes.



HOMEWORK

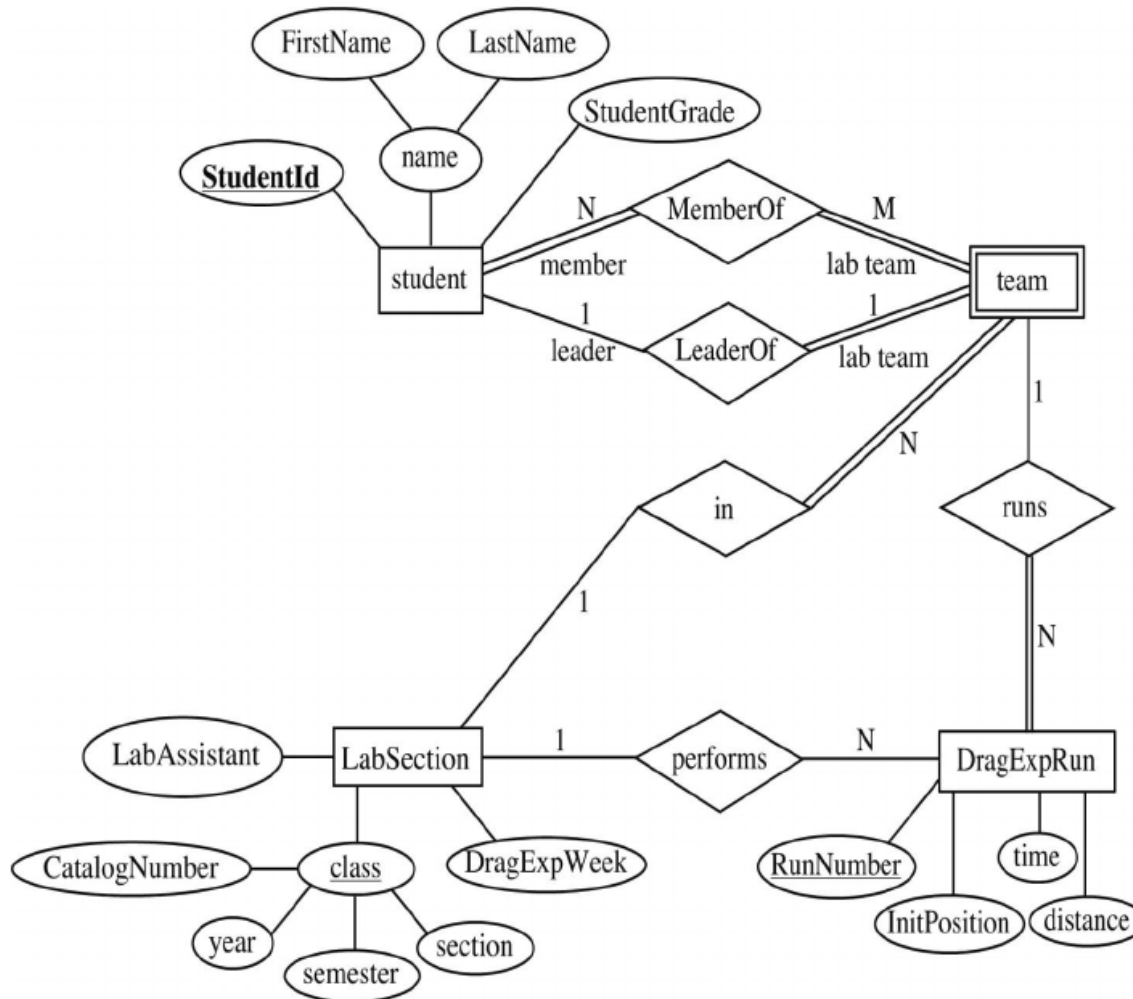
QUESTION 1:

For the diagram of ER Model below, construct the relational model using mapping from ER Model to Relational Model and show the relationships:



QUESTION 2:

For the diagram of ER Model below, construct the relational model using mapping from ER Model to Relational Model and show the relationships:



QUESTION 3:

For the given scenario below, draw the diagram of ER Model:

The Motor Vehicle Branch administers driving tests and issues driver's licenses. Any person who wants a driver's license must first take a learner's exam at any Motor Vehicle Branch in the province. If he/she fails the exam, he can take the exam again any time after a week of the failed exam date, at any branch. If he passes the exam, he is issued a license (type = learner's) with a unique license number. A learner's license may contain a single restriction on it. The person may take his driver's exam at any branch any time before the learner's license expiry date (which is usually set at six months after the license issuedate). If he passes the exam, the branch issues him a driver's license. A driver's license must also record if the driver has completed driver's education, for insurance purposes.

QUESTION 4:

For the given scenario below, draw the diagram of ER Model:

- Patients are identified by SSN, and their names, addresses, and also ages.
- Doctors are identified by an SSN, for each doctor, the name, specialty and years of experience must be recorded.
- Each pharmaceutical company is identified by name and has a phone number.
- For each drug, the trade name and formula must be recorded. Each drug is sold by a given pharmaceutical company, and the trade name identifies a drug uniquely from among the products of that company. If a pharmaceutical company is deleted, you need not keep track of its products any longer.
- Each pharmacy has a name, address, and phone number.
- Every patient has a primary physician. Every doctor has at least one patient.
- Each pharmacy sells several drugs and has a price for each. A drug could be sold at several pharmacies, and the price could vary from one pharmacy to another.
- Doctors prescribe drugs for patients. A doctor could prescribe one or more drugs for several patients, and a patient could obtain prescriptions from several doctors. Each prescription has a date and a quantity associated with it. You can assume that if a doctor prescribes the same drug for the same patient more than once, only the last such prescription needs to be stored.
- The pharmaceutical company has long-term contracts with pharmacies. A pharmaceutical company can contract with several pharmaceutical companies. For each contract, you have to store a start date, and end date, and the text of the contract.
- Pharmacies appoint a supervisor for each contract. There must always a supervisor for each contract.